Ankit Raj
6411390
Sogeti/Capgemini

🎓 Agentic AI Fundamentals Quiz

🧠 Section A: Multiple Choice Questions (MCQs)

1. What is Agentic AI?
A. AI that only performs predefined tasks
B. AI systems that act autonomously to achieve goals
C. AI used only for data storage
D. AI that requires constant human supervision

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2. Which of the following best describes an "agent" in AI?
A. A database
B. A program that perceives and acts upon an environment
C. A neural network only
D. A hardware device

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3. What is the primary goal of an AI agent?
A. To maximize randomness
B. To achieve a defined objective or goal
C. To reduce memory usage
D. To store large datasets

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4. Which component is responsible for deciding actions in an agent?
A. Sensor
B. Actuator
C. Policy/Decision-making module
D. Memory storage

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5. In Agentic AI, "planning" refers to:
A. Collecting data
B. Creating a sequence of actions to achieve a goal

C. Training a model
D. Visualizing results

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6. Which of the following is NOT a typical agent role?
A. Perception
B. Reasoning
C. Execution
D. Compilation

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7. Reasoning in AI agents helps to:
A. Store information
B. Make logical decisions based on data
C. Increase hardware speed
D. Reduce dataset size

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8. What does “execution” mean in Agentic AI?
A. Designing algorithms
B. Running planned actions in the environment
C. Training a model
D. Debugging code

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9. Which type of agent uses condition-action rules?
A. Learning agent
B. Model-based agent
C. Simple reflex agent
D. Utility-based agent

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10. Multi-agent systems involve:
A. Only one AI system
B. Multiple agents interacting with each other
C. Only human interaction
D. Only hardware devices

✍️ Section B: Short Answer Questions

11. Define Agentic AI in your own words.

Agentic AI is AI that can act independently to achieve goals by making decisions and taking actions.

12. List any three roles of an AI agent.

Perception, Reasoning, Execution

13. What is the difference between planning and execution?

Planning is deciding what steps to take, while execution is actually performing those steps.

14. Explain the importance of reasoning in AI agents.

Reasoning helps the agent make logical decisions based on available data and situations.

15. What are sensors and actuators in an agent system?

Sensors collect input from the environment, and actuators perform actions in the environment.

🔍 Section C: Application-Based Questions

16.
A delivery robot must pick up packages and deliver them efficiently.

- Identify the **agent roles** involved.

  Roles: Perception, Planning, Reasoning, Execution

17.
An AI chatbot analyzes user queries and responds intelligently.

- Explain how **reasoning** is used.

  The chatbot uses reasoning to understand the query and generate a meaningful and correct response.

18.
Design a simple agent for a **smart home system**.

- What would be its inputs, outputs, and goals?

  Inputs: Temperature, motion sensors

  Outputs: Turn lights/AC on or off

  Goals: Comfort, energy efficiency

19.
Explain how **planning + reasoning + execution** work together in a self-driving car.

Planning decides route, reasoning analyzes traffic and obstacles, execution controls the car movement.

20.
Why is autonomy important in Agentic AI? Give a real-world example.

Autonomy allows AI to work without constant human control. Example: Self-driving cars.

⚙️ Section D: Higher-Order Thinking

21. Compare Agentic AI with traditional rule-based systems.

Agentic AI is flexible and adapts to achieve goals in changing environments.
Rule-based systems follow fixed instructions and cannot adapt easily.

22. What challenges might arise in multi-agent systems?

Multi-agent systems may face coordination and communication issues.
Conflicts between agents can also reduce efficiency.

23. How can Agentic AI improve real-world decision-making?

Agentic AI improves decisions by analyzing data quickly and accurately.
It helps in making timely and efficient real-world choices.

24. Explain how feedback helps improve agent performance.

Feedback allows agents to learn from past actions and mistakes.
This helps improve their performance over time.

25. Can Agentic AI function without human input? Justify your answer.

Yes, Agentic AI can operate independently without constant human input.
However, humans are needed for initial setup and supervision.